Sleep Quality After Night Shifts Versus Days Off

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Research question

Null hypothesis : The mean sleep quality score is the same after night shifts and on days off.

Alternative hypothesis : The mean sleep quality score differs between sleep after night shifts and sleep on days off.

```
library(ggplot2)
```

```
## Warning: package 'ggplot2' was built under R version 4.5.3
```

```
library(dplyr)
```

```
## Warning: package 'dplyr' was built under R version 4.5.3
```

```
##
## Attaching package: 'dplyr'
```

```
## The following objects are masked from 'package:stats':
##
##     filter, lag
```

```
## The following objects are masked from 'package:base':
##
##     intersect, setdiff, setequal, union
```

```
library(readxl)
```

```
## Warning: package 'readxl' was built under R version 4.5.3
```

```
sleep_data <- read_excel("sleep_quality_project/sleep_data.xlsx")

sleep_data <- sleep_data %>%
  mutate(`sleep score` = round(100 * `sleep score`))

sleep_data
```

```
## # A tibble: 22 x 6
##    date       `day type` `sleep score` `sleep (hrs)` `Restless (min)` `HR (bpm)`
##    <chr>      <chr>              <dbl>         <dbl>             <dbl>      <dbl>
##  1 2026-10-1~ Night shi~            62             5                15         60
##  2 2026-10-2~ Day off               92             6                32         54
##  3 2026-10-2~ Day off               91            10                25         60
##  4 2026-10-2~ Night shi~            81             7                21         59
##  5 2026-10-2~ Night shi~            91             5                22         63
##  6 2026-10-2~ Day off               88             8                14         63
##  7 2026-10-2~ Day off               87             8                10         62
##  8 2026-10-2~ Day off               76             7                24         62
##  9 2026-10-2~ Day off               55             5                11         61
## 10 2026-10-2~ Day off               75             6                23         64
## # i 12 more rows
```

The labels follow the original log. A night-shift observation should refer to the main sleep episode after that shift, even if the shift ends the next day.

Missing measurements

```
#  filter() and is.na().
missing_data <- sleep_data %>% filter(is.na(`sleep score`))
missing_data
```

```
## # A tibble: 2 x 6
##   date       `day type` `sleep score` `sleep (hrs)` `Restless (min)` `HR (bpm)`
##   <chr>      <chr>              <dbl>         <dbl>             <dbl>      <dbl>
## 1 2026-11-02~ Day off               NA            NA                NA         NA
## 2 2026-11-06~ Day off               NA            NA                NA         NA
```

```
analysis_data <- sleep_data %>% filter(!is.na(`sleep score`))
```

Summary table: sleep score by day type

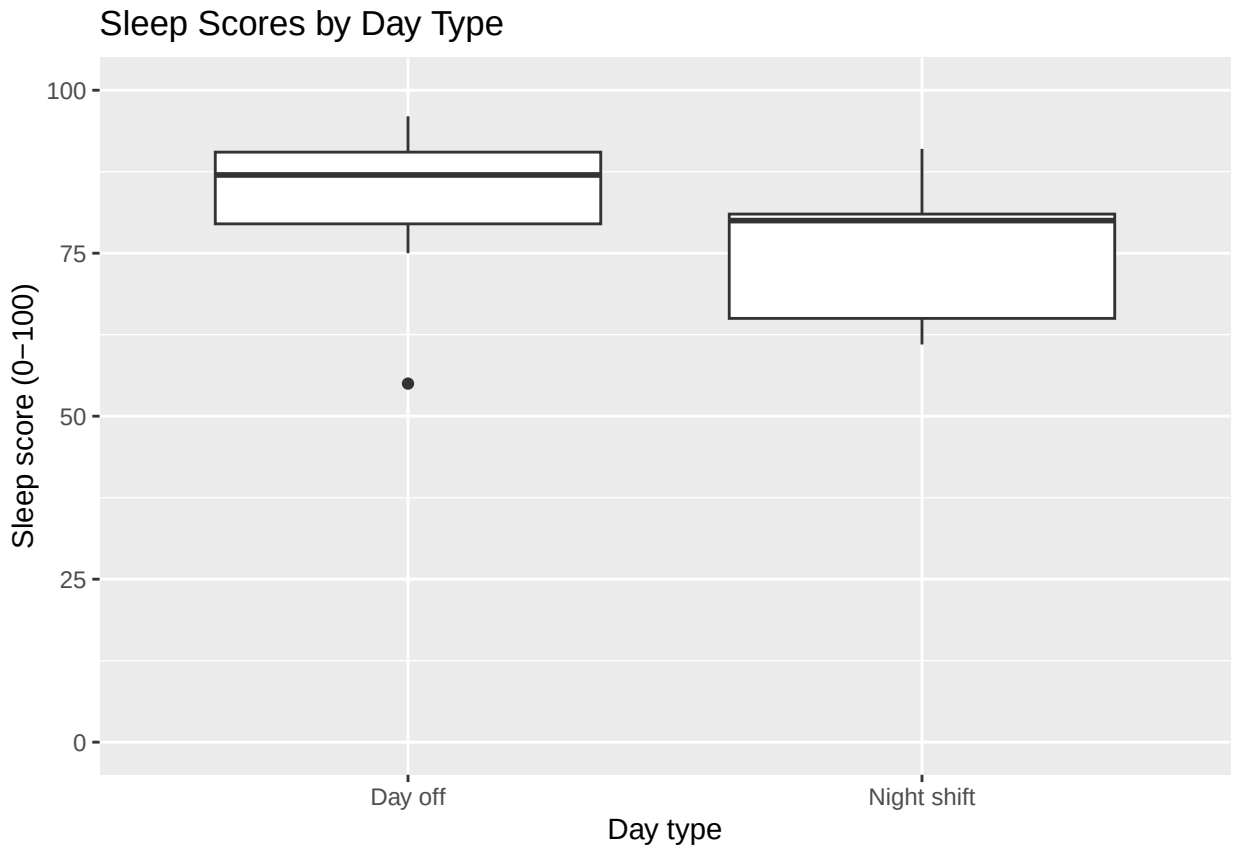
```
# : group_by(), summarize(), mean(), n().
summary_stats <- analysis_data %>%
  group_by(`day type`) %>%
  summarize(Count = n(), Mean = mean(`sleep score`),
            Minimum = min(`sleep score`), Maximum = max(`sleep score`))
summary_stats
```

```
## # A tibble: 2 x 5
##   `day type`  Count  Mean Minimum Maximum
##   <chr>       <int> <dbl>   <dbl>   <dbl>
## 1 Day off        11  83.4      55      96
## 2 Night shift     9  74.7      61      91
```

There are 9 scores after night shifts and 11 on days off. The mean scores are 74.67 and 83.36, respectively. Scores range from 61 to 91 after night shifts and from 55 to 96 on days off.

Graph 1: Sleep scores by day type

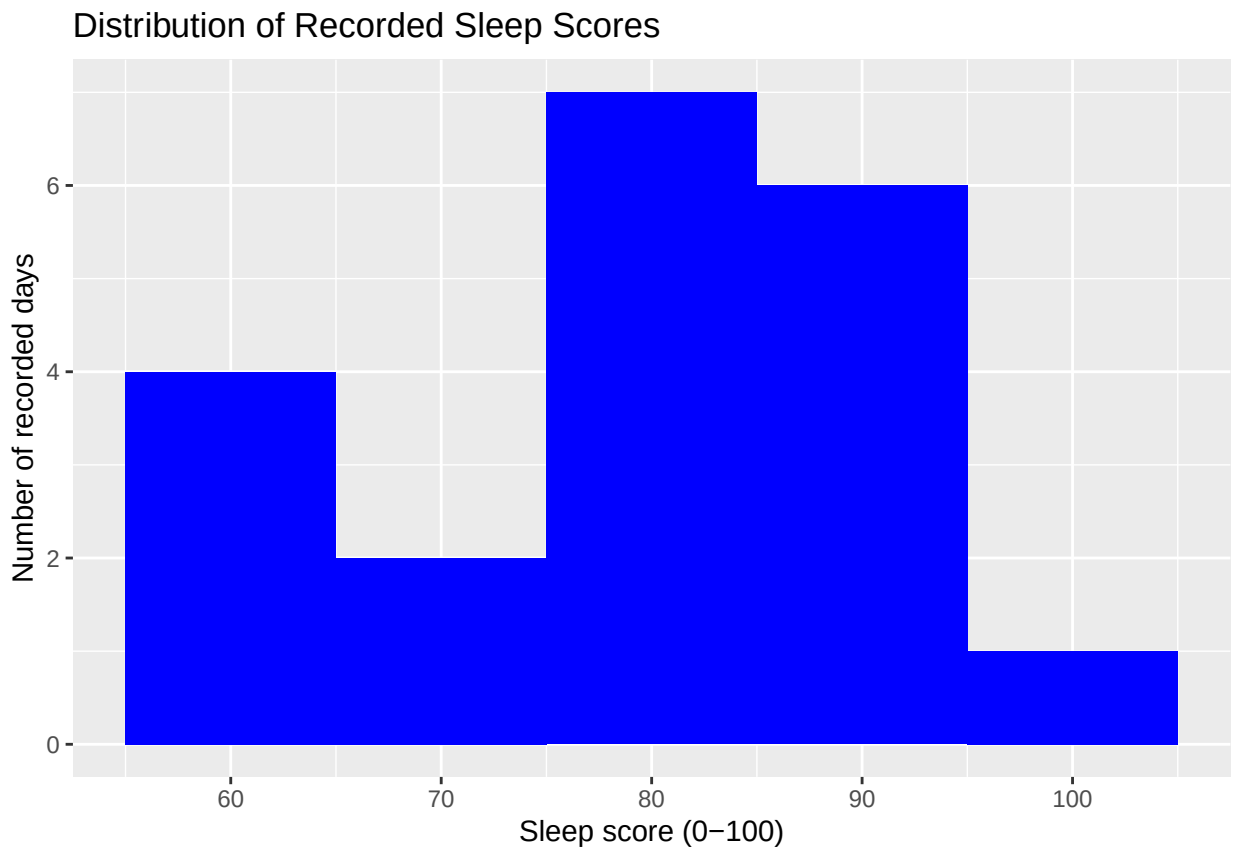
```
#  boxplot, limits, and labels.
ggplot(data = analysis_data, aes(x = `day type`, y = `sleep score`)) +
  geom_boxplot() +
  ylim(c(0, 100)) +
  labs(x = "Day type", y = "Sleep score (0-100)",
       title = "Sleep Scores by Day Type")
```



This graph compares the distributions of sleep scores in the two groups. A boxplot shows the median, spread, and possible outliers. The median is higher on days off (87 versus 80), but the groups overlap.

Graph 2: Distribution of sleep scores

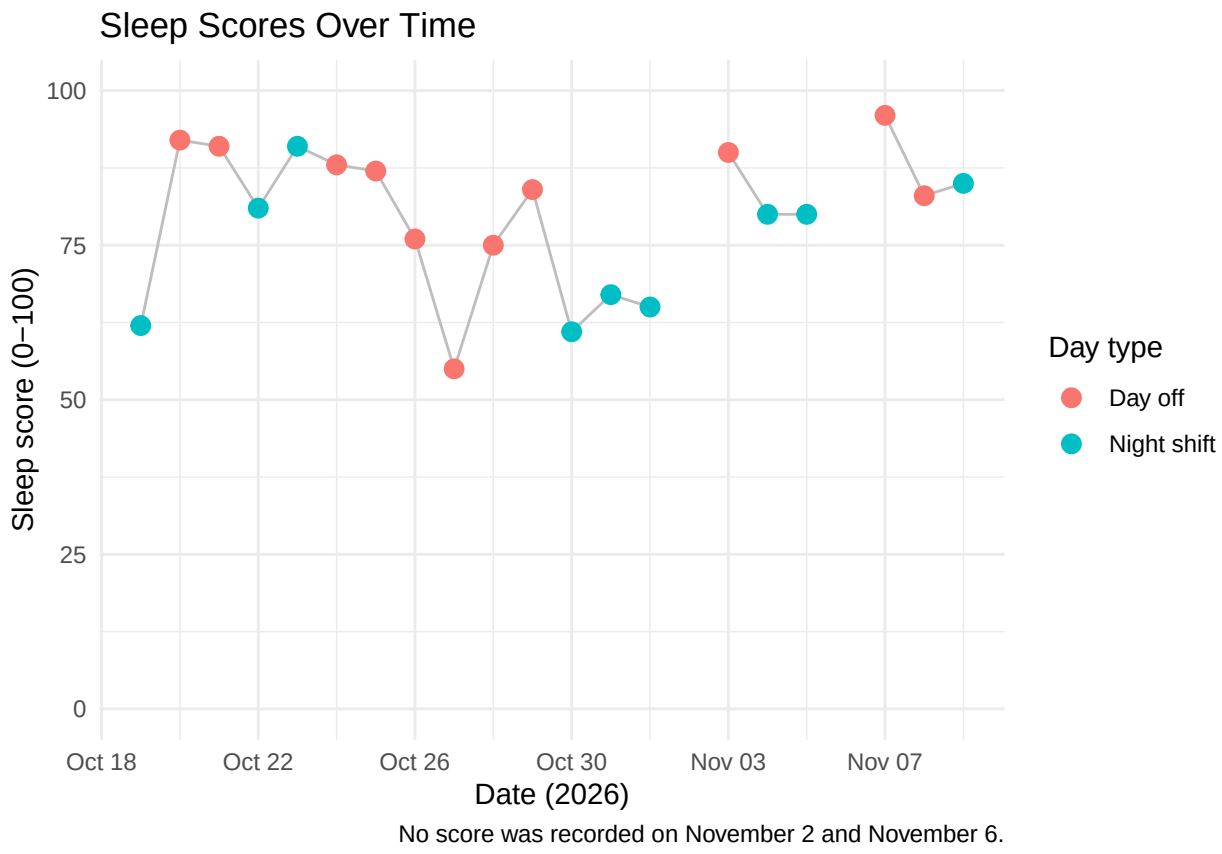
```
#  histogram with a chosen bin width.
ggplot(data = analysis_data, aes(x = `sleep score`)) +
  geom_histogram(binwidth = 10, fill = "blue") +
  labs(x = "Sleep score (0-100)", y = "Number of recorded days",
       title = "Distribution of Recorded Sleep Scores")
```



This histogram shows how scores are distributed across all recorded days. It groups nearby scores into intervals to show common and less common values. Most recorded scores are toward the higher end, with some lower scores.

```
## Graph 3: Sleep scores over time

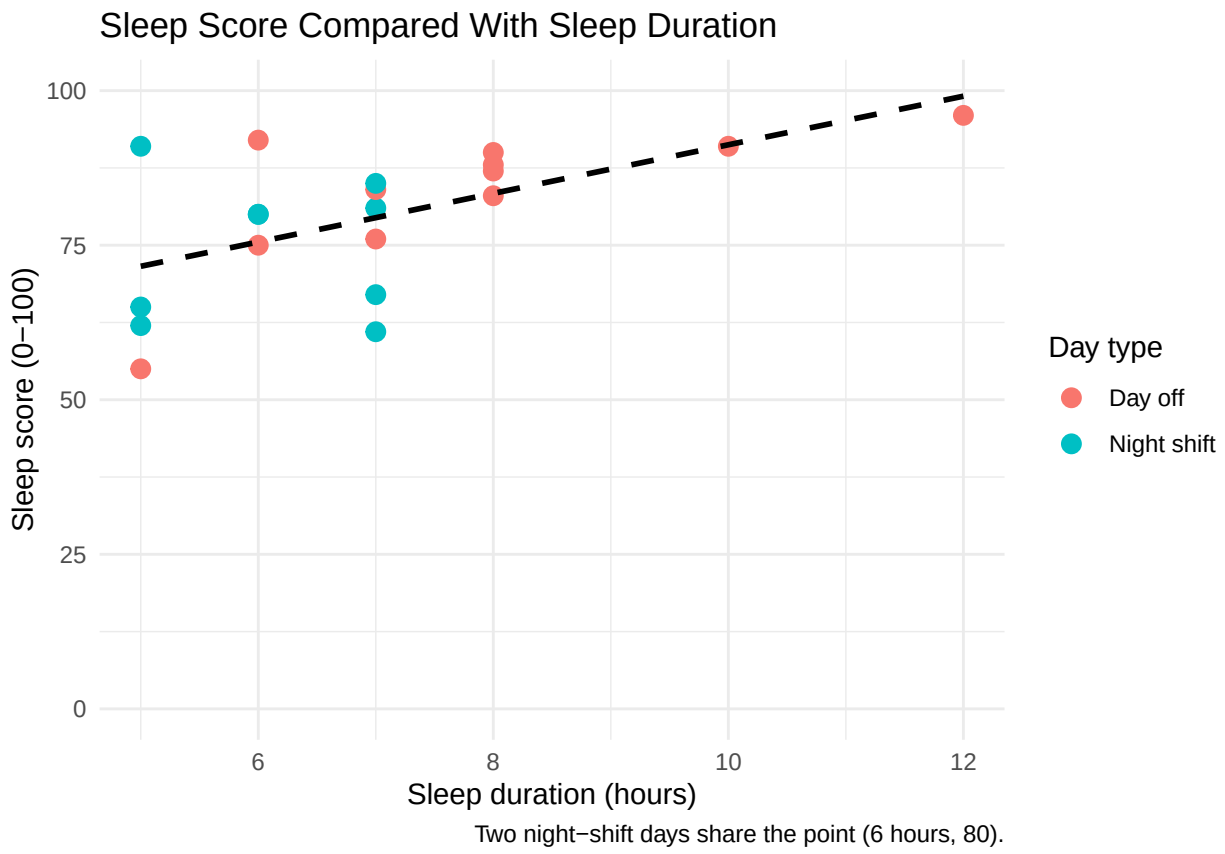
plot_time <- ggplot(data = sleep_data,
                    aes(x = as.Date(date), y = `sleep score`, group = 1)) +
  geom_line(color = "gray", na.rm = TRUE) +
  geom_point(aes(color = `day type`), size = 3, na.rm = TRUE) +
  scale_x_date(date_breaks = "4 days", date_labels = "%b %d") +
  ylim(c(0, 100)) +
  labs(x = "Date (2026)", y = "Sleep score (0-100)", color = "Day type",
       title = "Sleep Scores Over Time",
       caption = "No score was recorded on November 2 and November 6.") +
  theme_minimal()
print(plot_time)
```



Sleep score and sleep duration

```
# 7. GRAPH 4: SCATTER PLOT WITH REGRESSION LINE
# Each point is one recorded day: hours slept and that day's sleep score.
# geom_smooth(method = "lm") adds the regression line. se = FALSE hides
# the confidence band.
plot_scatter <- ggplot(data = analysis_data,
                       aes(x = `sleep (hrs)`, y = `sleep score`)) +
  geom_point(aes(color = `day type`), size = 3) +
  geom_smooth(method = "lm", se = FALSE, color = "black",
              linetype = "dashed") +
  ylim(c(0, 100)) +
  labs(x = "Sleep duration (hours)", y = "Sleep score (0-100)",
       color = "Day type",
       title = "Sleep Score Compared With Sleep Duration",
       caption = "Two night-shift days share the point (6 hours, 80).") +
  theme_minimal()
print(plot_scatter)
```

```
## `geom_smooth()` using formula = 'y ~ x'
```



Exploratory t-test: sleep score

```
# 8. TWO-SAMPLE T-TEST ON SLEEP SCORE
# mu1 = mean sleep score on night-shift days. mu2 = mean on days off.
# H0: mu1 = mu2. Ha: mu1 is not equal to mu2.
# filter() makes one data set for each group.
night_data <- analysis_data %>% filter(`day type` == "Night shift")
off_data <- analysis_data %>% filter(`day type` == "Day off")

# Night shift is first, so the difference is night shift minus day off.
# Compare the p-value with 0.05.
# Exploratory: repeated measurements from one person may not be independent.
score_test <- t.test(night_data$`sleep score`, off_data$`sleep score`,
                     alternative = "two.sided", var.equal = FALSE)
print(score_test)
```

```
##
##  Welch Two Sample t-test
##
## data:  night_data$‘sleep score‘ and off_data$‘sleep score‘
## t = -1.7292, df = 17.451, p-value = 0.1014
## alternative hypothesis: true difference in means is not equal to 0
## 95 percent confidence interval:
##  -19.287338   1.893399
```

```
## sample estimates:
## mean of x mean of y
##  74.66667  83.36364
```

The two-sided p-value is **0.1014**. At alpha = 0.05, fail to reject H0: this exploratory test provides insufficient evidence of a difference in mean sleep scores.

```
mean_difference <- mean(night_data$`sleep score`) - mean(off_data$`sleep score`)
reference_sd <- sqrt(
  (sd(night_data$`sleep score`)^2 + sd(off_data$`sleep score`)^2) / 2
)
cohens_d_unpooled <- mean_difference / reference_sd

effect_summary <- data.frame(
  Difference = mean_difference,
  Reference_SD = reference_sd,
  Unpooled_d = cohens_d_unpooled
)
knitr::kable(effect_summary, digits = 2, row.names = FALSE)
```

Difference	Reference_SD	Unpooled_d
-8.7	11.21	-0.78